ISHIKA MAHADAR

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Summary

Computer Engineering undergraduate (B.Tech, 2028) with hands-on experience building AI-agent, full-stack and computer-vision systems, from LLM agents and ML models to real-time dashboards and geospatial APIs. Skilled in Python, TypeScript, FastAPI, Next.js, React and PostgreSQL, with projects deployed as live demos across hackathons and research work. Two IEEE-conference papers, Meta OpenEnv Hackathon finalist, **300+** LeetCode problems solved.

Education

Vishwakarma Institute of Technology, Pune *Nov 2024 – Jun 2028 (expected)*
B.Tech in Computer Engineering CGPA: 8.96
Relevant Coursework: Data Structures, Theory of Computation, Operating Systems, Software Engineering, Computer Networks, Internet of Things

Projects

ATMC: Verified-Failure Memory for Coding Agents — GitHub *Oct 2026 – Present*
Python, Ollama, Qwen2.5-Coder, DeepSeek-Coder, Gemini, sandboxed code execution

- Built a self-improving coding assistant that stores sandbox-checker failures as notes and verified retries as worked examples, learning a team's hidden conventions with no weight updates on a laptop-run local model.
- Wrote a 110-question hidden-convention benchmark with a held-out set and rule-change streams; learning from verified failures lifted first-attempt pass rate by **+22.1 points** (95% CI 17.0–27.3) on Qwen2.5-Coder-7B and **+16.4** on DeepSeek-Coder-6.7B.
- Added a bounded rule memory with execution-based staleness detection that recovered **+23 to +26 points** over plain memory after conventions changed, storing **62–78% fewer notes**; ran pre-registered experiments and reported the ideas that failed.

Universal AI Payment Recovery Platform — GitHub — Live Demo *Sep 2026*
Next.js, TypeScript, Prisma, PostgreSQL, Vitest, Vercel (Razorpay AI Buildathon)

- Modeled recovery around the payment obligation instead of single transactions, so payments split across Razorpay and Stripe, partial payments, refunds, expired cards and provider outages resolve correctly instead of triggering redundant retries.
- Built a deterministic policy engine between the AI recovery agent and customer outreach, enforcing compliant escalation, opt-outs and stopping rules, with an audit trail and detection of rapid repeated card attempts.
- Deployed on Vercel with Postgres in Razorpay test mode; the dashboard computes recovered amount and AI-attributed recovery from live queries over a demo batch, backed by **118 passing Vitest tests** covering provider correlation, idempotency and webhooks.

Hiring Fleet: Multi-Agent AI Oversight — GitHub *Apr 2026*
Python, OpenEnv, GRPO, LoRA, Docker, Hugging Face Spaces

- Built a multi-agent environment where fraud, skills and timeline specialists investigate role-restricted resume sections and an Overseer agent decides from their reports alone, with role-gated action whitelists and siloed observations.
- Designed a deterministic reward of **7 independent sub-functions** with anti-exploit penalties over a **36-resume** dataset, needing no LLM judge.
- Fine-tuned an LLM with GRPO (LoRA), raising mean reward from **0.736 to 0.850** over 800 steps, and deployed it as a live Hugging Face Space.

Smart Garbage Detection — GitHub *Dec 2025 – Jan 2026*
YOLOv8, Flutter, FastAPI, PostgreSQL, PostGIS, Leaflet

- Trained a YOLOv8 model on the TACO dataset to classify waste, reaching **0.88 precision**, **0.85 recall** and **0.78 mAP@50**; published the work as a paper at IGNITE 2026 (IEEE).
- Built a Flutter app for GPS-tagged photo capture and a FastAPI backend that runs inference into PostgreSQL + PostGIS, averaging **1.8s** end to end.
- Sped up geospatial queries **30x** (420ms → 15ms) with GiST indexing, powering a Leaflet heatmap dashboard with automated municipal notifications.

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, SQL, Dart.
Backend & Data: FastAPI, Node.js/Express, Next.js, REST APIs, WebSockets, PostgreSQL/PostGIS, MongoDB, Prisma, FAISS, Qdrant.
AI/ML: LLM agents (Groq, Gemini, Claude APIs), GRPO/LoRA fine-tuning, Ollama, YOLOv8, multi-agent systems.
Frontend & Tools: React, Tailwind CSS, Flutter, Docker, Git, Vitest, Vercel, Chrome Extensions (Manifest V3).
Fundamentals: DSA, OOP, Operating Systems, Computer Networks, System Design (basics).

Publications & Achievements

- **Published** two IEEE conference papers: "Smart Assistance System for Alzheimer's Patients Using the Internet of Things" at **IEEE ICEI 2026** and "Smart Garbage Detection and Community Mapping Using YOLOv8" at **IGNITE 2026**.
- **Finalist (top 800 of 31,000 teams)** – Meta PyTorch OpenEnv Hackathon x Scaler School of Technology. *Apr 2026*
- **Top 2,800 of 1 lakh applicants** – Women Who Master Hackathon 2026 (Logitech x Aspire For Her), Zonal Round. *Sep 2026*
- Participant, GirlScript Summer of Code 2026 (open-source program); **300+ problems** solved on LeetCode.

Certifications

Meta Back-End Developer Professional Certificate (Coursera) — Fundamentals of Docker & Kubernetes (Scaler) — Exploring IoT with Cisco Packet Tracer (Cisco Networking Academy)